

CS549.V Midterm #2

I. QUESTION (6 POINTS)

- Give two synonym words.
- Give two similar but not synonym words.
- Give two related but not similar words.

II. QUESTION (10 POINTS)

What is the cosine-similarity between the vectors $\mathbf{v}$ and $\mathbf{v}$? Prove.

III. QUESTION (15 POINTS)

Given the following documents:

Doc1: ali topu at

Doc2: ali ayşeye çiçek al

Doc3: ali topu sev

Doc4: ayşe topu at

Doc5: ali ayşeye topu at

Calculate tf-idf weighted term-document matrix.

IV. QUESTION (5 POINTS)

Give examples for the following relations between senses:

- Antonymy
- Hypernymy
- Meronymy
- Instance-Hypernymy
- Derivation

V. QUESTION (15 POINTS)

Write the pseudocode for Lesk Word Sense Disambiguation algorithm.

VI. QUESTION (12 POINTS)

Given the following grammar L_1 below:

$$\begin{aligned}
 S &\rightarrow NP VP \\
 NP &\rightarrow Pronoun \\
 NP &\rightarrow ProperNoun \\
 NP &\rightarrow Det Nominal \\
 Nominal &\rightarrow Nominal Noun \\
 Nominal &\rightarrow Noun \\
 VP &\rightarrow Verb \\
 VP &\rightarrow Verb NP \\
 VP &\rightarrow Verb NP PP \\
 VP &\rightarrow Verb PP \\
 VP &\rightarrow VP PP \\
 PP &\rightarrow Prep NP \\
 Noun &\rightarrow \dots \\
 Verb &\rightarrow \dots \\
 Pronoun &\rightarrow \dots \\
 ProperNoun &\rightarrow \dots \\
 Det &\rightarrow \dots \\
 Prep &\rightarrow \dots
 \end{aligned}$$

- Fill the spaces with appropriate entries to create a grammar for which a parse tree for "The beauty is in the beholders eye" can be constructed.
- Construct the parse tree for that parse tree in a.

VII. QUESTION (12 POINTS)

Given the following grammar L_1 below:

$$\begin{aligned}
 S &\rightarrow NP VP \\
 NP &\rightarrow Pronoun \\
 NP &\rightarrow ProperNoun \\
 NP &\rightarrow Det Nominal \\
 Nominal &\rightarrow Nominal Noun \\
 Nominal &\rightarrow Noun \\
 VP &\rightarrow Verb \\
 VP &\rightarrow Verb NP \\
 VP &\rightarrow Verb NP PP \\
 VP &\rightarrow Verb PP \\
 VP &\rightarrow VP PP \\
 PP &\rightarrow Prep NP \\
 Noun &\rightarrow flights, breeze, trip, morning \\
 Verb &\rightarrow is, prefer, like, need, want, fly \\
 Pronoun &\rightarrow me, I, you, it \\
 ProperNoun &\rightarrow Alaska, Baltimore, Chicago, United \\
 Det &\rightarrow the, a, an, this, these, that \\
 Prep &\rightarrow from, to, on, near
 \end{aligned}$$

Convert the grammar into CNF form.

VIII. QUESTION (10 POINTS)

Given the following sentences, annotate the corresponding universal dependency relations on them.

Sentence	Relation
United canceled the flight	NSUBJ
United diverted the flight to Miami	DOBJ
We booked her the flight to Miami	IOBJ
We took the morning flight	NMOD
Book the cheapest flight	AMOD
JetBlue canceled 1000 flights	NUMMOD
The flight was canceled	DET
Book the flight through Houston	CASE
We flew Denver and Houston	CC, CONJ

IX. QUESTION (15 POINTS)

Given the following grammar L_2 in CNF form:

$$\begin{aligned}
 S &\rightarrow NP VP \\
 NP &\rightarrow Det Nominal \\
 Nominal &\rightarrow Noun Noun \\
 VP &\rightarrow Verb NP \\
 VP &\rightarrow Verb PP \\
 VP &\rightarrow VP PP \\
 PP &\rightarrow Prep NP \\
 Noun &\rightarrow flight, trip, super, morning \\
 Verb &\rightarrow is \\
 Det &\rightarrow the, this \\
 Prep &\rightarrow on
 \end{aligned}$$

Parse the sentence "This super trip is on the morning flight" using CKY parsing. Show your work.